

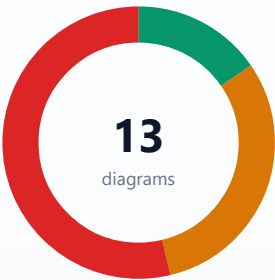
Architecture Diagrams

Pipeline: **brownfield** · Profile: **core** · State: **current** · Theme: **professional** · Generated: 2026-05-05T04:08:12.536Z

13 architecture diagrams derived from **11** evidence sources.

Every diagram is graph-derived, schema-validated, and carries an explicit confidence level. No generative model draws diagrams directly.

[docker](#) [node](#) [playwright](#) [postgres](#) [redis](#) [render](#)



- Confirmed **2** (15%)
- Inferred **0** (0%)
- Weak **4** (31%)
- Unknown **7** (54%)

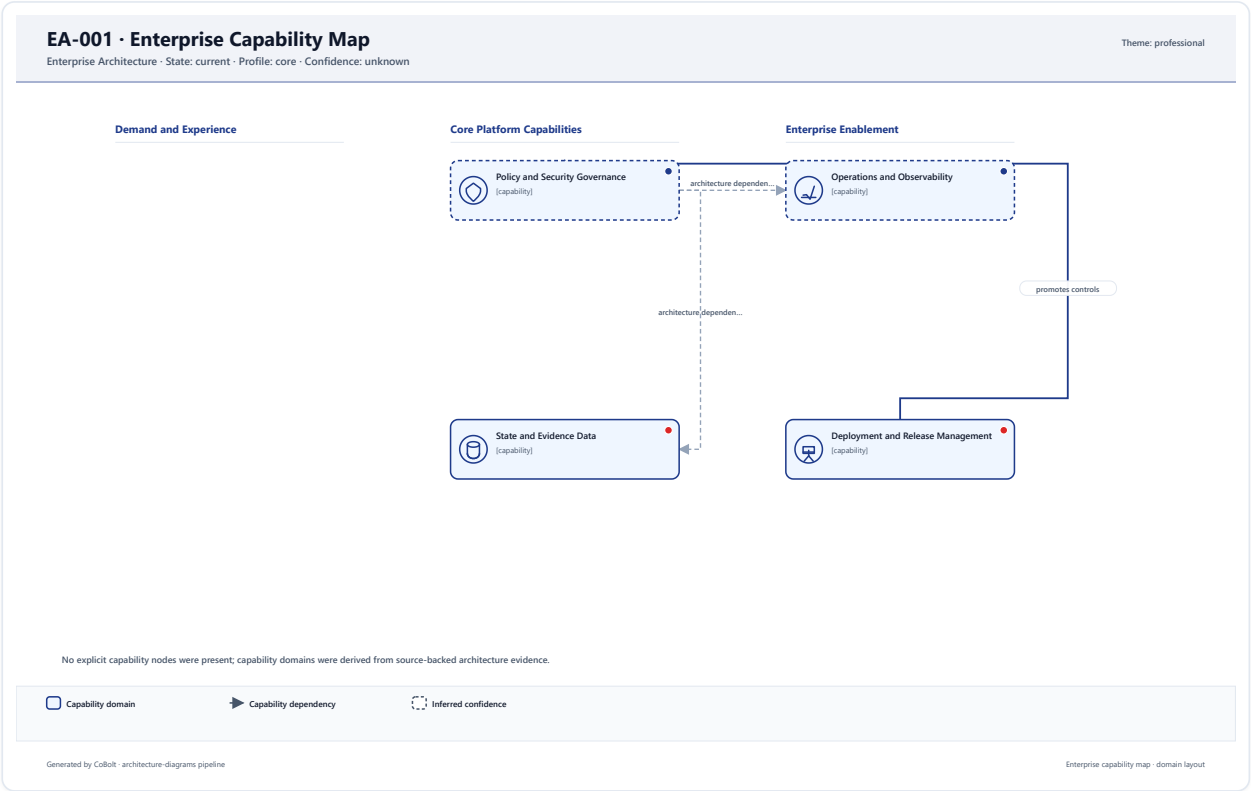
CONTENTS

Diagram Catalog	Integration Architecture (1)
Enterprise Architecture (1)	Platform Architecture (1)
Business Architecture (1)	Security Architecture (2)
Solution Architecture (2)	Operational Architecture (1)
Application Architecture (2)	Gap Report
Data Architecture (2)	

Diagram Catalog

ID	TITLE	AREA	STATE	CONF.	NODES	STATUS
EA-001	Enterprise Capability Map	Enterprise Architecture	CURRENT	UNKNOWN	4	generated
BA-001	Business Process / Value Stream	Business Architecture	CURRENT	CONFIRMED	2	generated
SA-001	Solution Architecture - System Context	Solution Architecture	CURRENT	CONFIRMED	4	generated
SA-002	Solution Architecture - Detailed Components	Solution Architecture	CURRENT	UNKNOWN	6	generated
APP-001	Application Architecture - Detailed Components	Application Architecture	CURRENT	UNKNOWN	13	generated
APP-002	Domain / Bounded Context Map	Application Architecture	CURRENT	WEAK	2	generated
DATA-001	Logical Data Model (ERD)	Data Architecture	CURRENT	UNKNOWN	1	generated
DATA-002	Data Flow Diagram	Data Architecture	CURRENT	UNKNOWN	1	generated
INT-001	Integration / API Map	Integration Architecture	CURRENT	WEAK	2	generated
PLAT-001	Runtime Platform Topology	Platform Architecture	CURRENT	UNKNOWN	4	generated
SEC-001	Security Trust Boundaries	Security Architecture	CURRENT	WEAK	4	generated
SEC-002	Identity and Access Flow	Security Architecture	CURRENT	UNKNOWN	0	warning
OPS-001	Observability and Operations Map	Operational Architecture	CURRENT	WEAK	2	generated

Nodes: 4 · Edges: 3 · Evidence refs: 7



Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

- package.json:@playwright/test — Detected "playwright" via package.json:@playwright/test
- docker-compose.yml — Service image matched "postgres"
- docker-compose.yml — Service image matched "redis"
- docker-compose.yml — Service image matched "docker"
- _cobolt-output/latest/planning/trd.md — External service "docker"
- _cobolt-output/latest/planning/trd.md — External service "render"

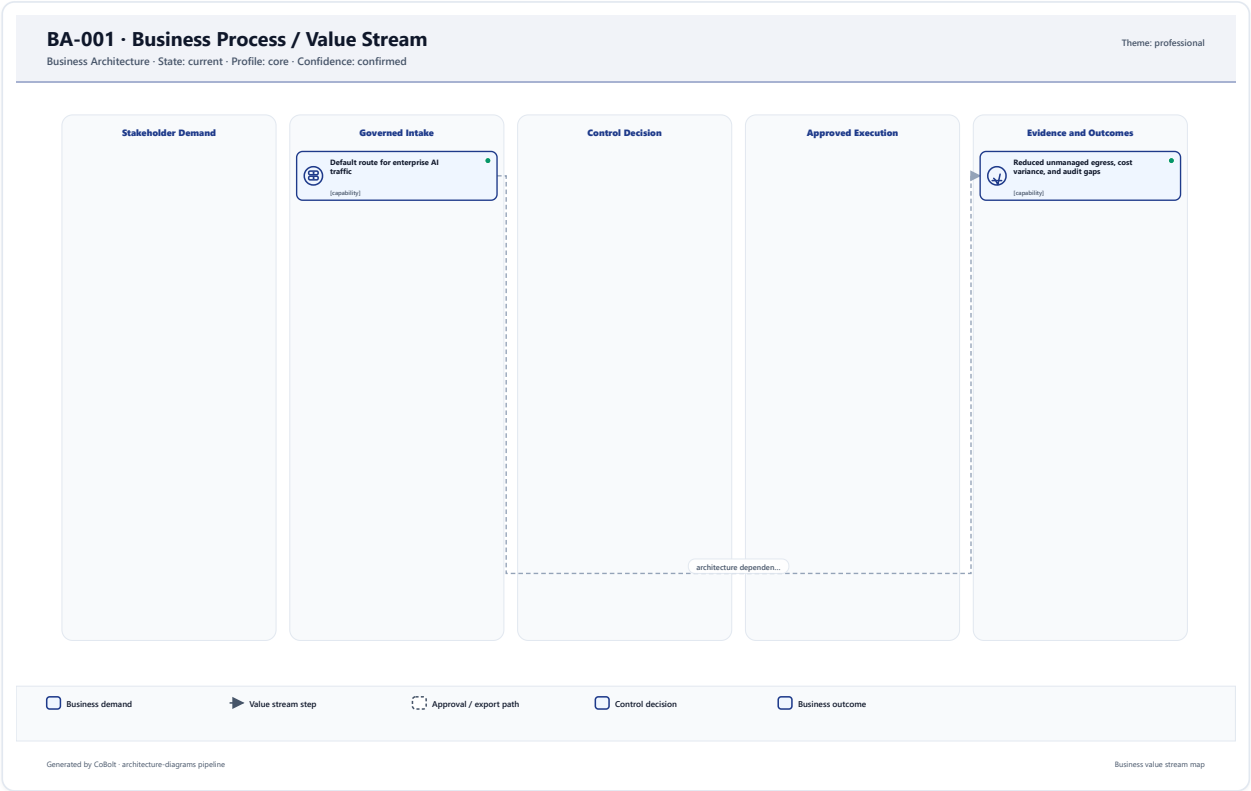
BA-001

Business Process / Value Stream

CURRENT

CONFIRMED

Nodes: 2 · Edges: 1 · Evidence refs: 2



Master spec

Mermaid

C4 / PlantUML

D2

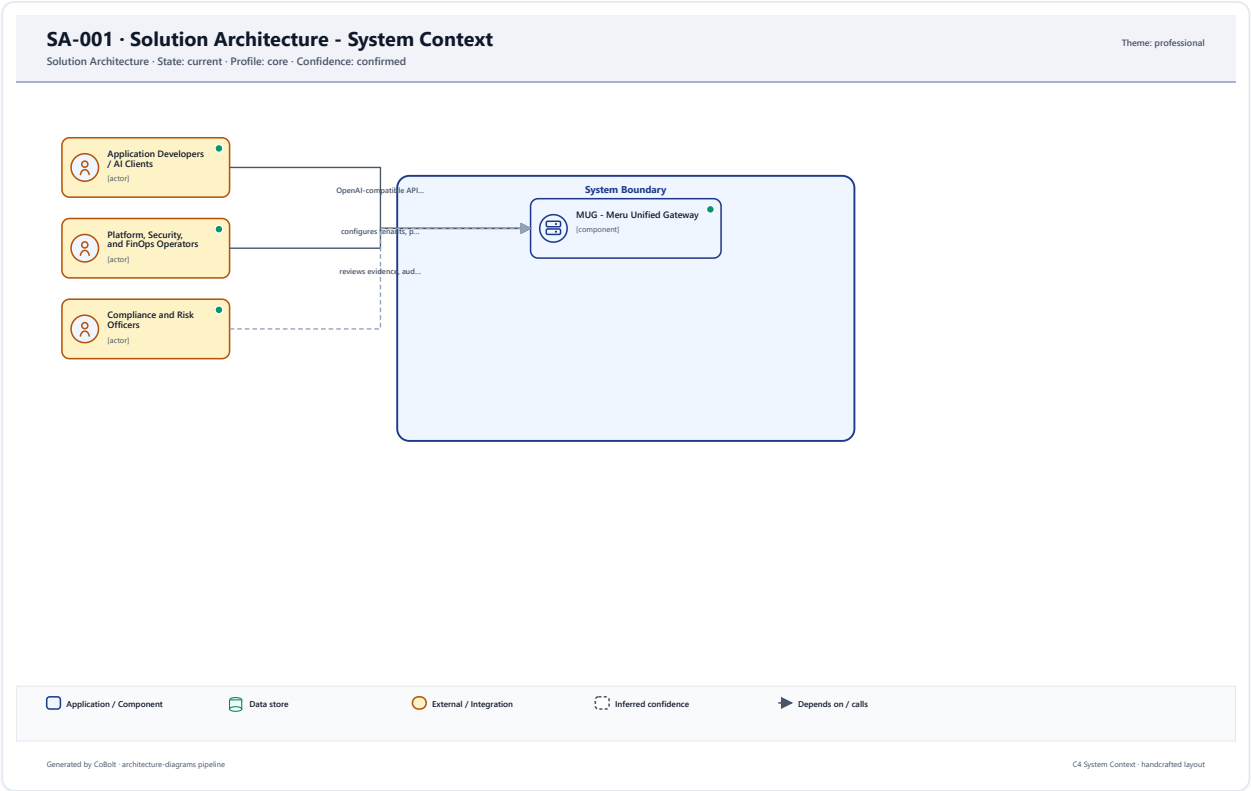
SVG iconic

Rendered SVG

Evidence

- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Business objective: MUG becomes the default route for enterprise AI traffic.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Business outcomes include reduced unmanaged AI use, lower cost variance, and improved audit readiness.

Nodes: 4 · Edges: 3 · Evidence refs: 4



Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

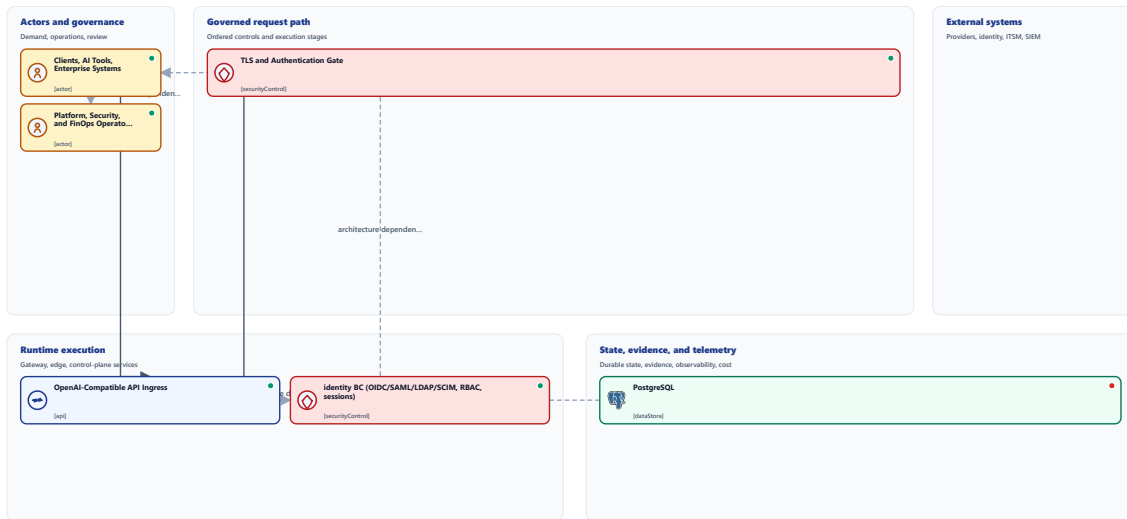
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Primary consumers send OpenAI-compatible and agent traffic into MUG.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Operators configure policies, budgets, approvals, and runtime controls.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Compliance stakeholders review evidence bundles and audit trails.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — MUG is the central governed gateway for enterprise AI traffic.

Nodes: 6 · Edges: 5 · Evidence refs: 6

SA-002 · Solution Architecture - Detailed Components

Solution Architecture · State: current · Profile: core · Confidence: unknown

Theme: professional



□ Actor / operator ▶ Primary request path ⋯ Escalation / export path 📄 State or evidence store ○ External system

Generated by Collob: architecture-diagrams pipeline

Solution Architecture - detailed components

Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived Clients, AI Tools, Enterprise Systems for the Solution Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived Platform, Security, and FinOps Operators for the Solution Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived OpenAI-Compatible API Ingress for the Solution Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived TLS and Authentication Gate for the Solution Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived identity BC (OIDC/SAML/LDAP/SCIM, RBAC, sessions) for the Solution Architecture viewpoint.
- `docker-compose.yml` — Service image matched "postgres"

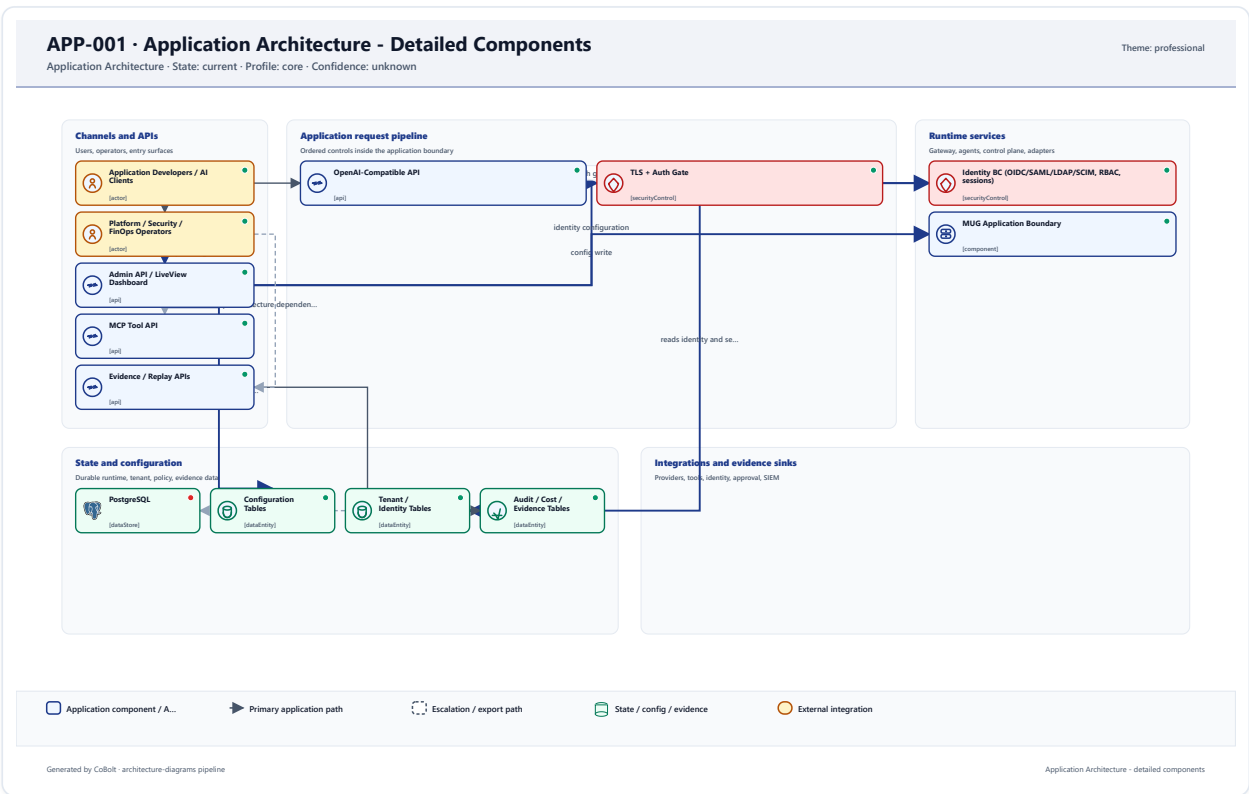
Application Architecture

APP-001 Application Architecture - Detailed Components

CURRENT

UNKNOWN

Nodes: 13 · Edges: 12 · Evidence refs: 13



Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

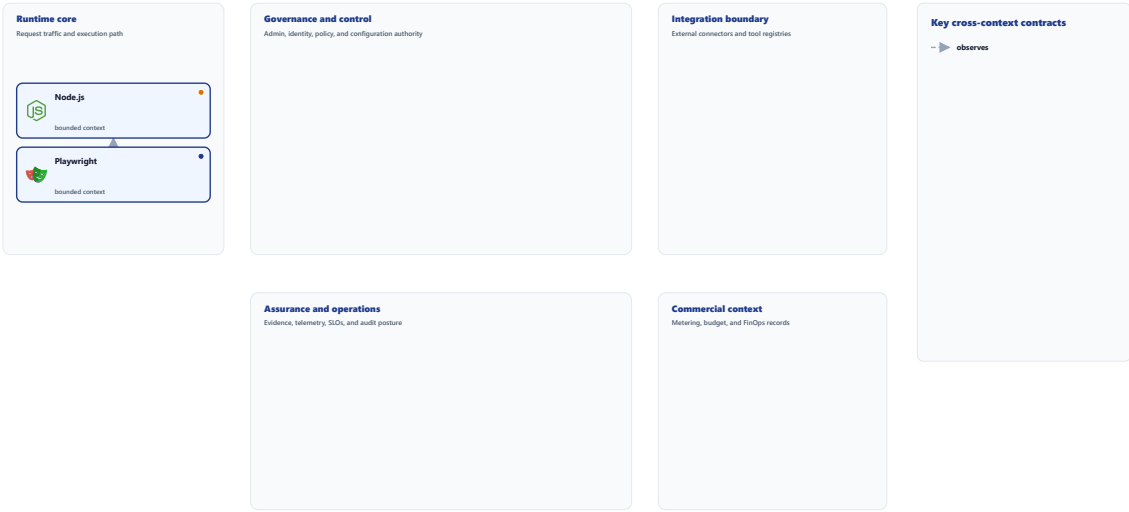
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived Application Developers / AI Clients for the Application Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Derived Platform / Security / FinOps Operators for the Application Architecture viewpoint.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — OpenAI-compatible request surface for application and agent traffic.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Administrative API and dashboard configure tenants, policy, routing, and controls.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — MCP-compatible surface accepts governed tool invocation traffic.
- `_cobolt-output/latest/brownfield/27-modernization-system-architecture.md` — Evidence, audit export, replay, and compliance-facing API surface.

Nodes: 2 · Edges: 1 · Evidence refs: 1

APP-002 · Domain / Bounded Context Map

Application Architecture · State: current · Profile: core · Confidence: weak

Theme: professional



Generated by Collob: architecture-diagrams pipeline

Domain / Bounded Context Map

Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

- package.json — Dependency declared as "node"

DATA-001

Logical Data Model (ERD)

CURRENT

UNKNOWN

Nodes: 1 · Edges: 0 · Evidence refs: 1

DATA-001 · Logical Data Model (ERD)

Data Architecture · State: current · Profile: core · Confidence: unknown

Theme: professional

Tenant and Identity

Logical data entities and relationships

Runtime Configuration

Logical data entities and relationships

Policy and Guardrails

Logical data entities and relationships

Provider and Tool Catalog

Logical data entities and relationships

Evidence and Operations

Logical data entities and relationships

Storage Substrate

Logical data entities and relationships

Postgres System of Record

[system-of-record]



Data entity / durable sto...



Runtime or policy control



Primary relationship / fl...



Projection / export / der...



External provider or data...

Generated by Colbolt - architecture-diagrams pipeline

Data Architecture - Logical Data Model (ERD)

Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

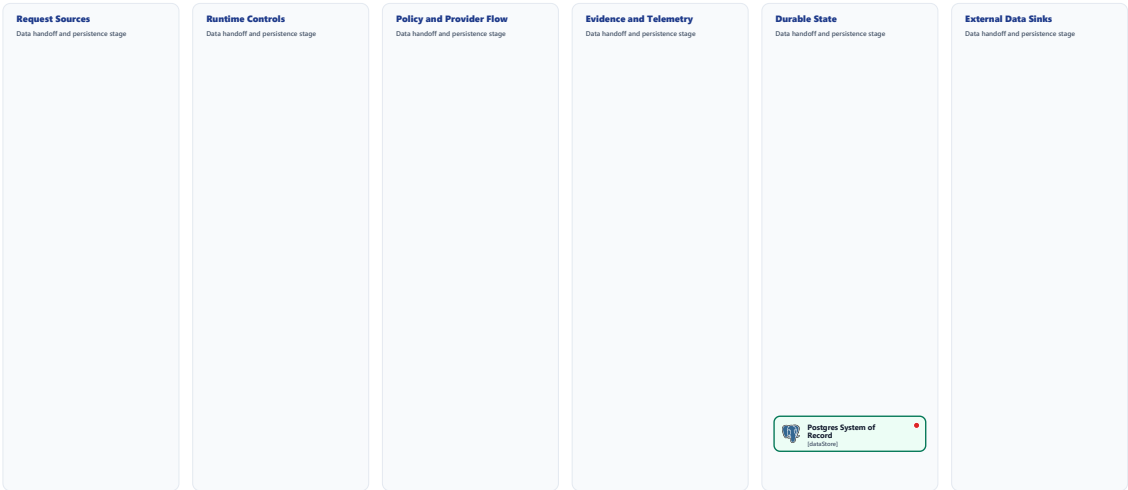
- docker-compose.yml — Service image matched "postgres"

Nodes: 1 · Edges: 0 · Evidence refs: 1

DATA-002 · Data Flow Diagram

Data Architecture · State: current · Profile: core · Confidence: unknown

Theme: professional



Data entity / durable sto...

Runtime or policy control

Primary relationship / fl...

Projection / export / det...

External provider or data...

Generated by Collob: architecture-diagrams pipeline

Data Architecture - Data Flow Diagram

Master spec

Mermaid

C4 / PlantUML

D2

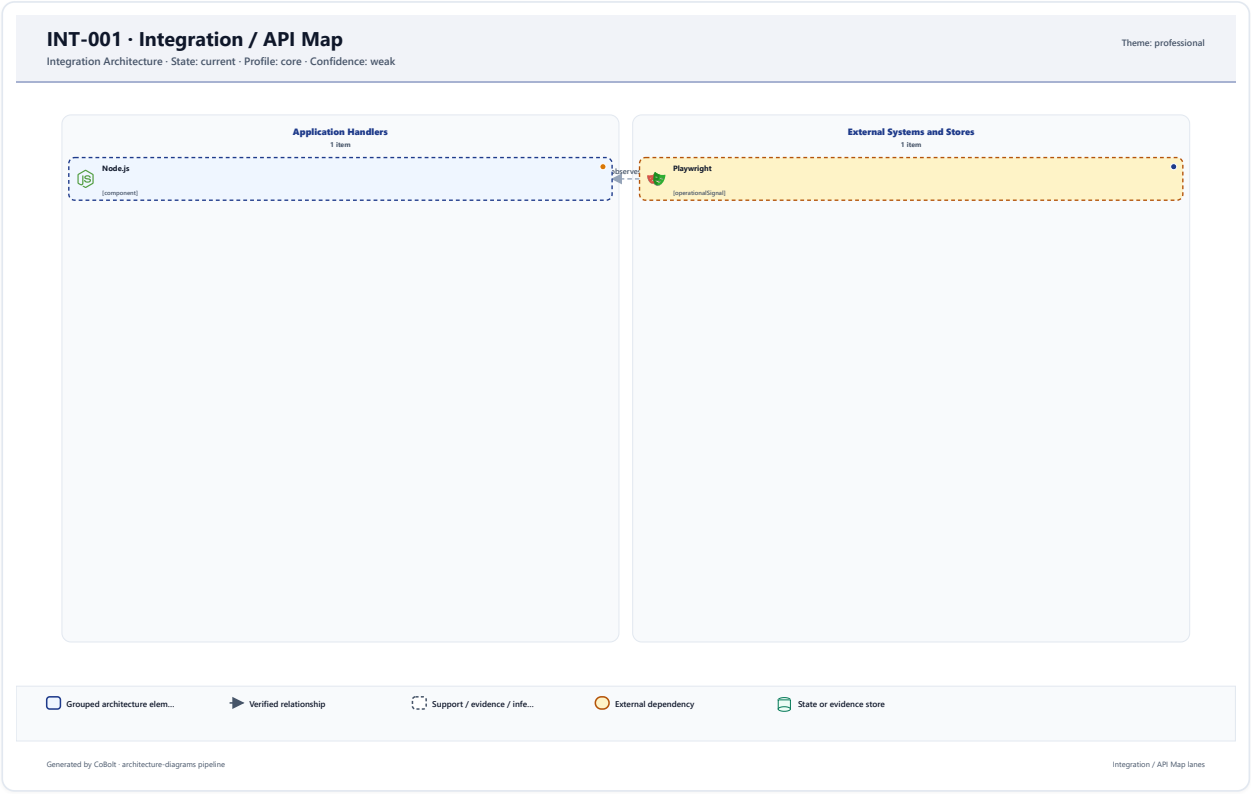
SVG iconic

Rendered SVG

Evidence

- docker-compose.yml — Service image matched "postgres"

Nodes: 2 · Edges: 1 · Evidence refs: 1



PLAT-001

Runtime Platform Topology

CURRENT

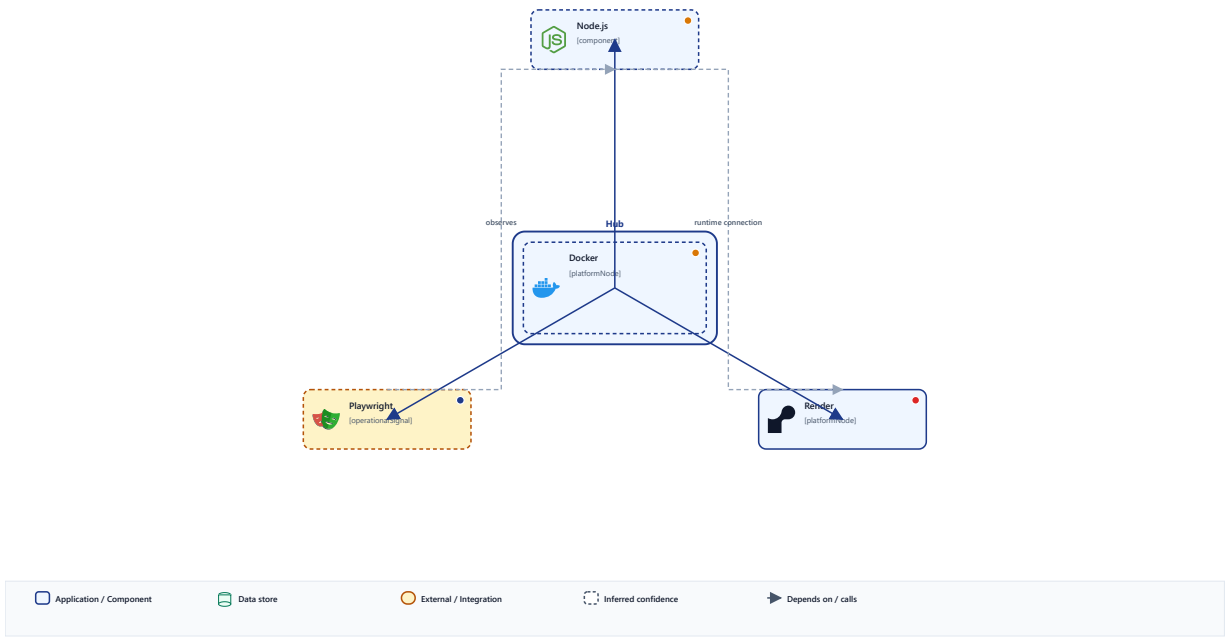
UNKNOWN

Nodes: 4 · Edges: 3 · Evidence refs: 5

PLAT-001 · Runtime Platform Topology

Platform Architecture · State: current · Profile: core · Confidence: unknown

Theme: professional



Generated by Cobolt - architecture-diagrams pipeline

Hub-and-spoke topology

Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

- package.json — Dependency declared as "node"
- docker-compose.yml — Service image matched "docker"
- _cobolt-output/latest/planning/trd.md — External service "docker"
- _cobolt-output/latest/planning/trd.md — External service "render"
- _cobolt-output/latest/planning/prd.md — Mentioned "render"

SEC-001

Security Trust Boundaries

CURRENT

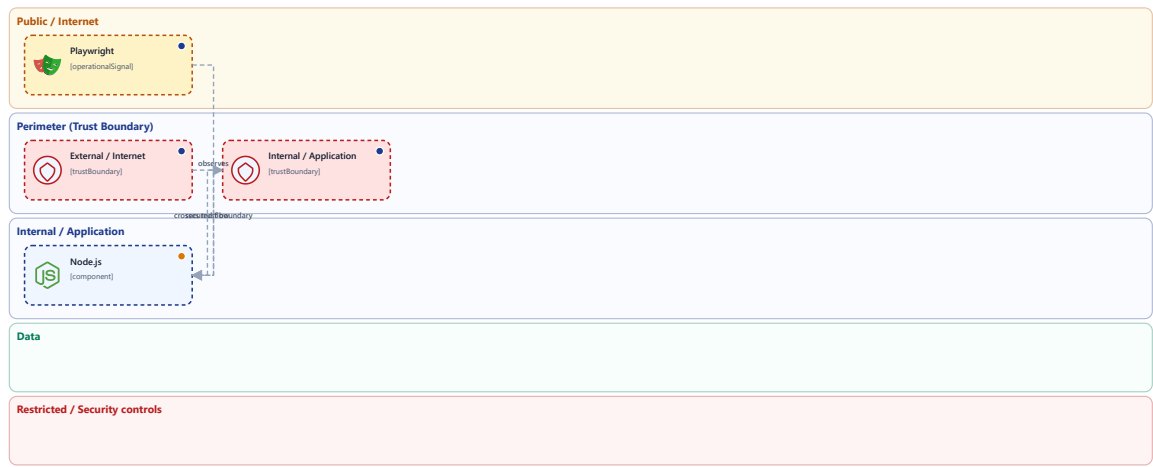
WEAK

Nodes: 4 · Edges: 3 · Evidence refs: 1

SEC-001 · Security Trust Boundaries

Security Architecture · State: current · Profile: core · Confidence: weak

Theme: professional



Public Internal Data Trust boundary Controlled flow

Generated by Colbolt - architecture diagrams pipeline

Security trust zones — handcrafted layout

Master spec

Mermaid

C4 / PlantUML

D2

SVG iconic

Rendered SVG

Evidence

- package.json — Dependency declared as "node"

SEC-002

Identity and Access Flow

CURRENT

UNKNOWN

Nodes: 0 · Edges: 0 · Evidence refs: 0 · Note: No graph nodes matched viewpoint

SEC-002 · Identity and Access Flow

Security Architecture · State: current · Profile: core · Confidence: unknown

Theme: professional

No graph nodes matched this viewpoint.

Application / Component

Data store

External / Integration

Inferred confidence

Depends on / calls

Generated by Collobi - architecture-diagrams pipeline

Master spec

Mermaid

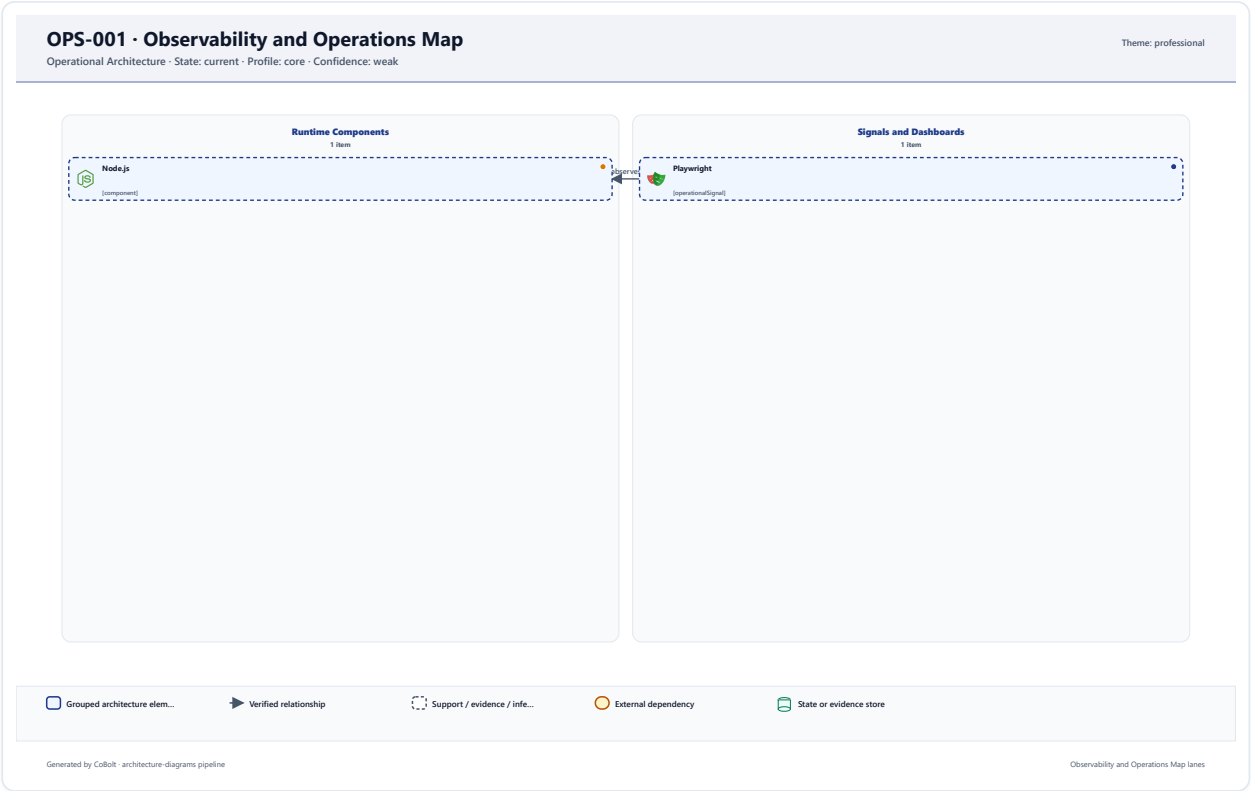
C4 / PlantUML

D2

SVG iconic

Rendered SVG

Nodes: 2 · Edges: 1 · Evidence refs: 2



Gap Report

Confidence summary — confirmed: **6**, inferred: **2**, weak: **0**, unknown: **0**

actor — No actor nodes extracted from available evidence.

capability — No capability nodes extracted from available evidence.

dataEntity — No dataEntity nodes extracted from available evidence.

api — No api nodes extracted from available evidence.

integration — No integration nodes extracted from available evidence.

securityControl — No securityControl nodes extracted from available evidence.

actor — Add actor 'End User (macOS local)' — primary and only human actor for a desktop viewer.

capability — Add capabilities: render-markdown, render-mermaid-diagrams, render-katex-math, syntax-highlight, sanitize-html, parse-frontmatter, pan-zoom-svg, file-open.

component — Add component-electrobun (runtime shell), component-bun (JS runtime, replacing inferred component-node), component-markdown-it (+ 6 plugins), component-mermaid, component-katex, component-highlight-js, component-dompurify, component-gray-matter, component-svg-pan-zoom — all evidenced in package.json dependencies.

dataEntity — Add dataEntity 'MarkdownDocument' (file path + parsed AST + frontmatter) and 'DesignTokens' (referenced via scripts/design-token-check.ts). No relational entities apply.

integration — No external integrations exist; mark N/A explicitly. Remove inferred Postgres/Redis/Render which are scaffolding phantoms.

securityControl — Add securityControl-dompurify (HTML sanitisation), securityControl-katex-trust-false, securityControl-mermaid-security-level, securityControl-electrobun-renderer-sandbox, securityControl-bun-audit (CI gate via 'bun audit --audit-level=high').

operationalSignal — Reclassify ops-playwright as test-tooling component, not operational signal. Add real ops signals only if found (none currently in package.json).

platformNode — Replace plat-docker/plat-render with plat-macos (target platform, evidenced by 'Native macOS markdown viewer' in package.json description and gen-icons output) and plat-bun-runtime.

scaffolding-cleanup — Remove or quarantine entity-postgres, entity-redis, plat-docker, plat-render — all four are CoBolt _cobolt-docker / planning-template phantoms with zero evidence in actual project source (package.json, scripts/, e2e/).

